
BENCHMARK APPENDIX

Benchmark Appendix

Validation methodology, baselines, and results

Aurelius

AI infrastructure orchestration and cost optimization

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Purpose

This appendix is the technical validation reference. It describes how Aurelius is benchmarked, the controls that keep the results honest, the baselines it is measured against, and the limitations of the current validation.

What is measured

The benchmark measures the energy-cost reduction Aurelius achieves on a workload relative to a baseline scheduling policy, using real wholesale electricity prices. The headline metric is percentage cost reduction versus a current-price-only baseline, reported per workload class and as a mean across classes. Secondary metrics — carbon impact, SLA violations, migration count, downside events, and per-fold variance — are reported alongside.

Data

Market	Region	Signal
CAISO	US West	Day-ahead and real-time prices
PJM	US East	Day-ahead and real-time prices
ERCOT	US South	Day-ahead and real-time prices

Validation uses two real historical windows: Q1 2026 (higher winter volatility) and Summer 2025 (more stable), with no missing price hours in the reported configurations. All economic claims use real, unrandomized data from the source markets; synthetic and sandbox data are structurally barred from any savings path.

Walk-forward validation

Evaluation is leakage-free walk-forward, not in-sample. For each fold the forecaster is trained only on price data preceding the fold's evaluation window, then evaluated on the held-out window. The validated configuration uses 30-day training windows and five walk-forward folds. The same temporal split is applied to every optional signal, so a future reading cannot influence a past decision.

Leakage prevention

Leakage prevention is treated as a correctness property and is independently tested. Forecaster training data is filtered to the pre-evaluation window per fold; real-time settlement prices are never visible to the forecaster or optimizer at decision time; optional signals use a last-known-before-decision lookup; and synthetic provenance raises an error if it reaches a benchmark path.

During development, a predict-time feature was inadvertently zero-filling the future window, manufacturing an artificial signal that inflated apparent savings. The defect was found in review, the inflated result was discarded and never

archived, and the corrected figure is what is reported. The harness is designed so that this class of error surfaces rather than ships.

Baselines

Baseline	Definition	Role
Current-price-only	Always route to the cheapest region at submission, using live prices	Primary — strong, realistic
Upper-bound diagnostic	Optimizer with perfect future price knowledge	Diagnostic ceiling, not a baseline
Naive single-region	No optimization	Reference only; not used for claims

The headline metric is always versus the current-price-only baseline. It is the strongest realistic baseline — it has perfect current-price information and represents what a sophisticated manual operator achieves — so improvement over it is attributable to forecasting, cross-region routing, and rescheduling rather than to a weak comparison.

Results summary

Window	Configuration	Mean reduction
Q1 2026 (CAISO / PJM / ERCOT)	Quantile forecaster, 30-day windows, 5 folds	25.0%
Summer 2025 (CAISO / PJM / ERCOT)	Quantile forecaster, 30-day windows, 5–7 folds	22.8%

Savings persist across both seasonal windows, which is the relevant robustness check. Per-workload figures appear in the ROI methodology and the Executive Summary.

Upper-bound diagnostic

The upper-bound diagnostic quantifies how much of the available opportunity the optimizer captures. A small gap means the optimizer is near the structural limit; a large gap means forecasting is the binding constraint.

Workload	Validated (p50)	Upper-bound diagnostic	Gap
LLM batch inference	~34%	~43%	~9 pts
Training	~15%	~30%	~15 pts
Fine-tuning	~13%	~47%	~33 pts

For batch inference the optimizer is near-optimal. For training and fine-tuning the larger gaps are driven by winter price volatility that the forecaster only partly anticipates; on stable summer data these gaps narrow to single digits. This is disclosed because it bounds the realistic upside and explains the variation between seasons.

Limitations

The current validation is bounded, and the bounds are stated rather than implied.

- Coverage is three U.S. markets; European and Asia-Pacific markets are not yet validated.
- Two seasonal windows of roughly 90 days each are validated; full-year persistence is not yet demonstrated.
- Published figures use workloads generated from per-class profiles; customer-specific results require the customer's own trace, confirmed in shadow mode.
- Carbon-aware results are limited to CAISO marginal-emissions coverage on the available data plan.
- Queue-aware and GPU-health signals have been validated only against synthetic fixtures, which are excluded from savings claims.
- Experimental forecaster variants did not consistently beat the production model in the windows tested and are not the default.